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compact1, compact2, compact3

java.io

Class FileOutputStream

java.lang.Object
    java.io.OutputStream
        java.io.FileOutputStream

All Implemented Interfaces:

`Closeable`, `Flushable`, `AutoCloseable`

```
public class FileOutputStream
extends OutputStream
```

A file output stream is an output stream for writing data to a `File` or to a `FileDescriptor`. Whether or not a file is available or may be created depends upon the underlying platform. Some platforms, in particular, allow a file to be opened for writing by only one `FileOutputStream` (or other file-writing object) at a time. In such situations the constructors in this class will fail if the file involved is already open.

`FileOutputStream` is meant for writing streams of raw bytes such as image data. For writing streams of characters, consider using `FileWriter`.

Since:

`JDK1.0`

See Also:

`File`, `FileDescriptor`, `FileInputStream`, `Files.newOutputStream(java.nio.file.Path, java.nio.file.OpenOption...)`

Constructor Summary

Constructors

Constructor	Description
`FileOutputStream(File file)`	Creates a file output stream to write to the file represented by the specified `File` object.
`FileOutputStream(File file, boolean append)`	Creates a file output stream to write to the file represented by the specified `File` object.
`FileOutputStream(FileDescriptor fdObj)`	Creates a file output stream to write to the specified file descriptor, which represents an existing connection to an actual file in the file system.
`FileOutputStream(String name)`	Creates a file output stream to write to the file with the specified name.
`FileOutputStream(String name, boolean append)`	Creates a file output stream to write to the file with the specified name.

Method Summary

Concrete Methods

Modifier and Type	Method	Description
`void`	`close()`	Closes this file output stream and releases any system resources associated with this stream.
`protected void`	`finalize()`	Cleans up the connection to the file, and ensures that the `close` method of this file output stream is called when there are no more references to this stream.

FileChannel	getChannel()	Returns the unique FileChannel object associated with this file output stream.
FileDescriptor	getFD()	Returns the file descriptor associated with this stream.
void	write(byte[] b)	Writes b.length bytes from the specified byte array to this file output stream.
void	write(byte[] b, int off, int len)	Writes len bytes from the specified byte array starting at offset off to this file output stream.
void	write(int b)	Writes the specified byte to this file output stream.

Methods inherited from class java.io.OutputStream

flush

Methods inherited from class java.lang.Object

clone, equals, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructor Detail

FileOutputStream

```
public FileOutputStream(String name)
                 throws FileNotFoundException
```

Creates a file output stream to write to the file with the specified name. A new FileDescriptor object is created to represent this file connection.

First, if there is a security manager, its checkWrite method is called with name as its argument.

If the file exists but is a directory rather than a regular file, does not exist but cannot be created, or cannot be opened for any other reason then a FileNotFoundException is thrown.

Parameters:

name - the system-dependent filename

Throws:

FileNotFoundException - if the file exists but is a directory rather than a regular file, does not exist but cannot be created, or cannot be opened for any other reason

SecurityException - if a security manager exists and its checkWrite method denies write access to the file.

See Also:

SecurityManager.checkWrite(java.lang.String)

FileOutputStream

```
public FileOutputStream(String name,
                        boolean append)
                 throws FileNotFoundException
```

Creates a file output stream to write to the file with the specified name. If the second argument is `true`, then bytes will be written to the end of the file rather than the beginning. A new `FileDescriptor` object is created to represent this file connection.

First, if there is a security manager, its `checkWrite` method is called with `name` as its argument.

If the file exists but is a directory rather than a regular file, does not exist but cannot be created, or cannot be opened for any other reason then a `FileNotFoundException` is thrown.

Parameters:

name - the system-dependent file name

append - if true, then bytes will be written to the end of the file rather than the beginning

Throws:

FileNotFoundException - if the file exists but is a directory rather than a regular file, does not exist but cannot be created, or cannot be opened for any other reason.

SecurityException - if a security manager exists and its checkWrite method denies write access to the file.

Since:

JDK1.1

See Also:

SecurityManager.checkWrite(java.lang.String)

FileOutputStream

```
public FileOutputStream(File file)
                 throws FileNotFoundException
```

Creates a file output stream to write to the file represented by the specified `File` object. A new `FileDescriptor` object is created to represent this file connection.

First, if there is a security manager, its `checkWrite` method is called with the path represented by the `file` argument as its argument.

If the file exists but is a directory rather than a regular file, does not exist but cannot be created, or cannot be opened for any other reason then a `FileNotFoundException` is thrown.

Parameters:

`file - the file to be opened for writing.`

Throws:

`FileNotFoundException - if the file exists but is a directory rather than a regular file, does not exist but cannot be created, or cannot be opened for any other reason`

`SecurityException - if a security manager exists and its checkWrite method denies write access to the file.`

See Also:

`File.getPath()`, `SecurityException`, `SecurityManager.checkWrite(java.lang.String)`

FileOutputStream

```
public FileOutputStream(File file,
                        boolean append)
                 throws FileNotFoundException
```

Creates a file output stream to write to the file represented by the specified `File` object. If the second argument is `true`, then bytes will be written to the end of the file rather than the beginning. A new `FileDescriptor` object is created to represent this file connection.

First, if there is a security manager, its `checkWrite` method is called with the path represented by the `file` argument as its argument.

If the file exists but is a directory rather than a regular file, does not exist but cannot be created, or cannot be opened for any other reason then a `FileNotFoundException` is thrown.

Parameters:

file - the file to be opened for writing.

append - if true, then bytes will be written to the end of the file rather than the beginning

Throws:

FileNotFoundException - if the file exists but is a directory rather than a regular file, does not exist but cannot be created, or cannot be opened for any other reason

SecurityException - if a security manager exists and its checkWrite method denies write access to the file.

Since:

1.4

See Also:

File.getPath(), SecurityException, SecurityManager.checkWrite(java.lang.String)

FileOutputStream

```
public FileOutputStream(FileDescriptor fdObj)
```

Creates a file output stream to write to the specified file descriptor, which represents an existing connection to an actual file in the file system.

First, if there is a security manager, its `checkWrite` method is called with the file descriptor `fdObj` argument as its argument.

If `fdObj` is null then a `NullPointerException` is thrown.

This constructor does not throw an exception if `fdObj` is `invalid`. However, if the methods are invoked on the resulting stream to attempt I/O on the stream, an `IOException` is thrown.

Parameters:

fdObj - the file descriptor to be opened for writing

Throws:

SecurityException - if a security manager exists and its checkWrite method denies write access to the file descriptor

See Also:

SecurityManager.checkWrite(java.io.FileDescriptor)

Method Detail

write

```
public void write(int b)
           throws IOException
```

Writes the specified byte to this file output stream. Implements the `write` method of `OutputStream`.

Specified by:

`write` in class `OutputStream`

Parameters:

`b` - the byte to be written.

Throws:

`IOException` - if an I/O error occurs.

write

```
public void write(byte[] b)
           throws IOException
```

Writes `b.length` bytes from the specified byte array to this file output stream.

Overrides:

`write` in class `OutputStream`

Parameters:

`b` - the data.

Throws:

`IOException` - if an I/O error occurs.

See Also:

`OutputStream.write(byte[], int, int)`

write

```
public void write(byte[] b,
                  int off,
                  int len)
           throws IOException
```

Writes `len` bytes from the specified byte array starting at offset `off` to this file output stream.

Overrides:

`write` in class `OutputStream`

Parameters:

`b` - the data.

`off` - the start offset in the data.

`len` - the number of bytes to write.

Throws:

`IOException` - if an I/O error occurs.

close

```
public void close()
           throws IOException
```

Closes this file output stream and releases any system resources associated with this stream. This file output stream may no longer be used for writing bytes.

If this stream has an associated channel then the channel is closed as well.

Specified by:

`close` in interface `Closeable`

Specified by:

`close` in interface `AutoCloseable`

Overrides:

`close` in class `OutputStream`

Throws:

`IOException` - if an I/O error occurs.

getFD

```
public final FileDescriptor getFD()
                           throws IOException
```

Returns the file descriptor associated with this stream.

Returns:

the FileDescriptor object that represents the connection to the file in the file system being used by this FileOutputStream object.

Throws:

`IOException` - if an I/O error occurs.

See Also:

FileDescriptor

getChannel

```
public FileChannel getChannel()
```

Returns the unique FileChannel object associated with this file output stream.

The initial position of the returned channel will be equal to the number of bytes written to the file so far unless this stream is in append mode, in which case it will be equal to the size of the file. Writing bytes to this stream will increment the channel's position accordingly. Changing the channel's position, either explicitly or by writing, will change this stream's file position.

Returns:

the file channel associated with this file output stream

Since:

1.4

finalize

```
protected void finalize()
                throws IOException
```

Cleans up the connection to the file, and ensures that the close method of this file output stream is called when there are no more references to this stream.

Overrides:

finalize in class Object

Throws:

IOException - if an I/O error occurs.

See Also:

```
FileInputStream.close()
```


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